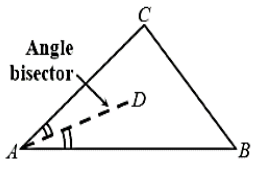
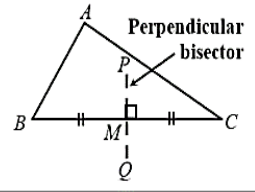
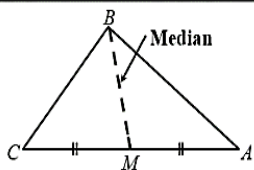
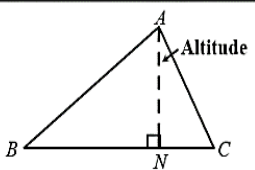


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Angle bisector <i>(A straight line which bisects an interior angle of a triangle)</i>	Perpendicular bisector <i>(A straight line which is perpendicular to and bisects one side of a triangle)</i>
	
Median <i>(A line segment drawn from a vertex of a triangle to the mid-point of its opposite side)</i>	Altitude <i>(A perpendicular line segment drawn from a vertex of a triangle to its opposite side)</i>
	

Properties of Angle bisector:

1. If a point lies on the angle bisector, then it is equidistant from the two sides of the angle.
2. If a point is equidistant from the two sides of an angle, then it lies on the angle bisector.

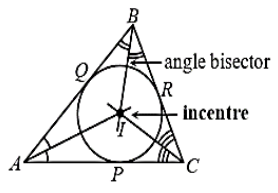
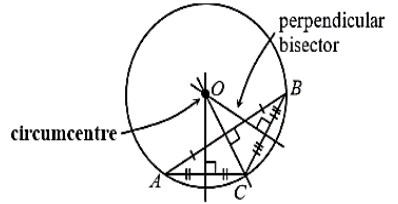
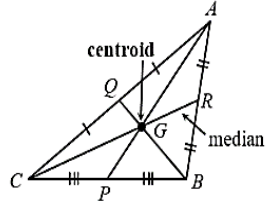
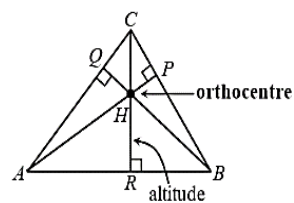
Properties of Perpendicular bisector:

1. If a point lies on the perpendicular bisector of a line segment, then it is equidistant from the two end points of the line segment
2. If a point is equidistant from the two end points of a line segment, then it lies on the perpendicular bisector of the line segment.

Concurrent:

-Lines in a plane are said to be **concurrent** if they intersect at a single point.

*In a triangle, four basic types of sets of concurrent lines are **altitudes**, **angle bisectors**, **medians** and **perpendicular bisectors**.

Incentre <i>(point of intersection of 3 angle bisectors)</i>	Circumcentre <i>(point of intersection of 3 perpendicular bisectors)</i>
	
Centroid <i>(point of intersection of 3 medians)</i>	Orthocentre <i>(point of intersection of 3 altitudes)</i>
	

Properties of Incentre:

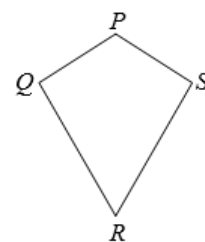
1. The incentre of a triangle is equidistant from the three sides of the triangle.
2. A circle inside the triangle can be constructed with this distance as its radius and the incentre as its centre.

Properties of Circumcentre:

1. The circumcentre of a triangle is equidistant from the three vertices of the triangle.
2. A circle passing through the vertices can be constructed with this distance as its radius and the circumcentre as its

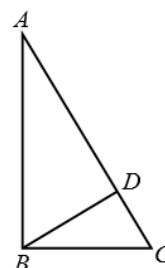
1. In the figure, $PQ = PS$ and $QR = RS$. Prove that $\angle PQR = \angle PSR$.

(3 marks)



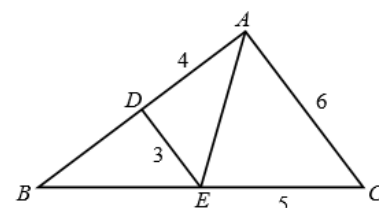
2. In the figure, ADC is a straight line. AB is an altitude of $\triangle ABC$ and $\angle BAD = \angle CBD$. Prove that BD is an altitude of $\triangle BCD$.

(2 marks)



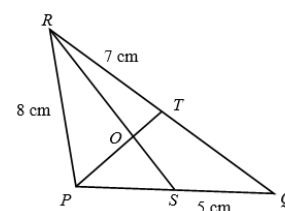
3. In the figure, ADB and BEC are straight lines. ED is the perpendicular bisector of AB . Using the properties of perpendicular bisectors, find the perimeter of $\triangle ABC$.

(3 marks)



4. In the figure, O is the centroid of $\triangle PQR$. PO and RO are produced to meet QR and PQ at T and S respectively. Find the perimeter of $\triangle PQR$.

(2 marks)



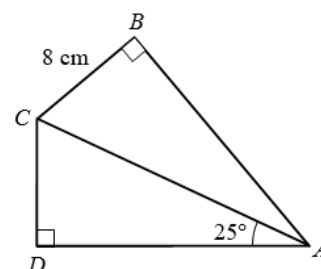
5. In the figure, AC is the angle bisector of $\angle BAD$.

(a) Find $\angle ACB$.

(2 marks)

(b) Using the properties of angle bisectors, find CD .

(1 mark)



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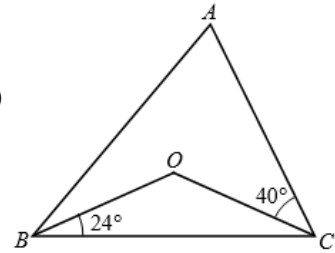
6. In the figure, O is the circumcentre of $\triangle ABC$.

(a) Find $\angle OCB$.

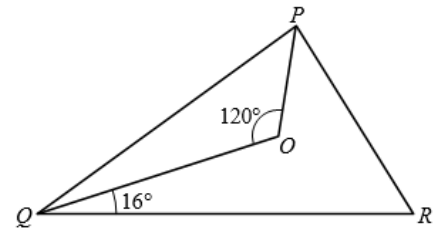
(b) Find $\angle BAC$.

(1 mark)

(3 marks)



7. In the figure, O is the incentre of $\triangle PQR$. $\angle POQ = 120^\circ$ and $\angle OQR = 16^\circ$. Find $\angle PRQ$. (2 marks)



8. In the figure, AD and BE are produced to meet at C . It is given that $\angle ABC = \angle DEC$.

(a) Prove that $\triangle ABC \sim \triangle DEC$.

(2 marks)

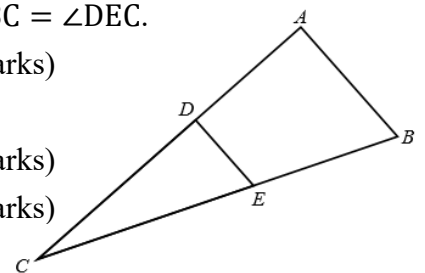
(b) If $CD = 36$ cm, $DA = 24$ cm, $CE = 39$ cm and $DE = 15$ cm,

(i) prove that $\triangle CDE$ is a right-angled triangle,

(2 marks)

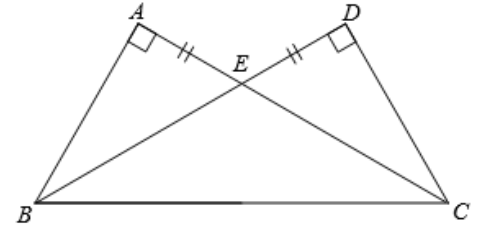
(ii) find the area of $\triangle ABC$.

(2 marks)

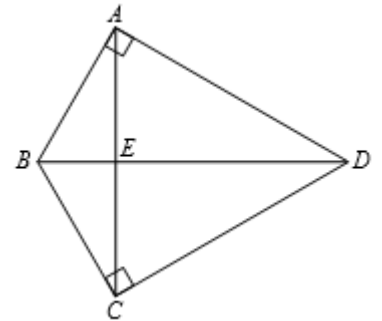


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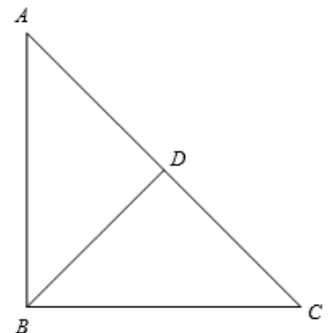
9. In the figure, BED and AEC are straight lines. EC is the angle bisector of $\angle DCB$. $EA = ED$ and $\angle BAE = \angle CDE = 90^\circ$. Is BE the angle bisector of $\angle ABC$? Explain your answer. (2 marks)



10. In the figure, AC intersects BD at E . $\angle DCB = \angle DAB = 90^\circ$. BD is the angle bisector of $\angle ABC$. Prove that BE is a perpendicular bisector of ΔABC . (2 marks)



11. In the figure, $AB = BC$. D is a point on AC such that the orthocentre of ΔABC lies on BD .
- Prove that $\Delta ABD \cong \Delta CBD$. (2 marks)
 - Does the circumcentre of ΔABC lie on BD ? Explain your answer. (1 mark)
 - Does the incentre of ΔABC lie on BD ? Explain your answer. (1 mark)



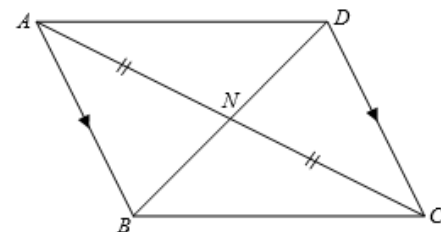
12. In the figure, AC and BD intersect at N. It is given that $AN = CN$ and $AB \parallel DC$.

(a) Prove that $\triangle ABN \cong \triangle CDN$.

(2 marks)

(b) Prove that $AD \parallel BC$.

(2 marks)



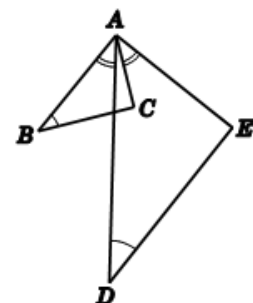
13. In the figure, $\angle ABC = \angle ADE$, $\angle BAD = \angle CAE$, $AB = 5$ cm, $AD = 10$ cm, $AE = 6$ cm and $DE = 8$ cm.

(a) Prove that $\triangle ABC \sim \triangle ADE$.

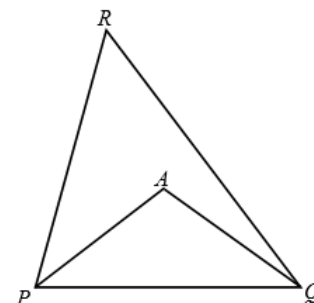
(2 marks)

(b) Find the area of $\triangle ABC$.

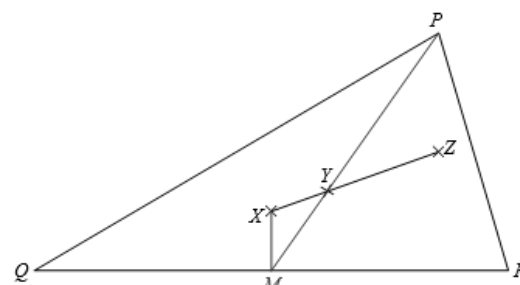
(3 marks)



14. In the figure, A is the circumcentre of $\triangle PQR$. $PQ = PR = 40$ and $QR = 48$. Find the area of $\triangle APQ$.
(4 marks)

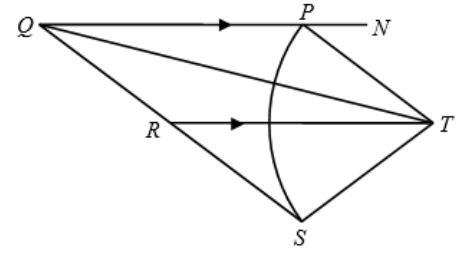


15. In the figure, Y and Z are the centroid and the orthocentre of $\triangle PQR$ respectively. PYM and XYZ are straight lines. $MY : YP = XY : YZ = 1 : 2$. Prove that XM is a perpendicular bisector of $\triangle PQR$. (3 marks)

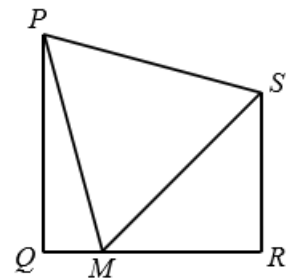


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16. In the figure, $\triangle TPS$ is a sector such that T is a centre. If $PQ \parallel TR$ and TR is an angle bisector of $\triangle TPS$. Prove that TR is a median of $\triangle QTS$. (3 marks)

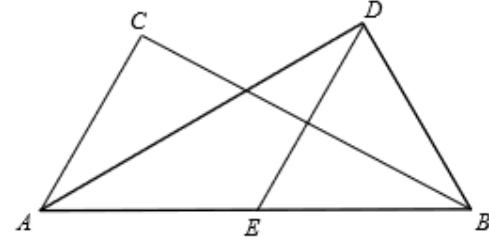


17. In the figure, $\angle PQR = \angle SRQ = 90^\circ$ and M is a point on QR such that $\triangle PMS$ is an equilateral triangle. If $PQ = QR$ and $SR = 6$, find the perimeter of $\triangle PMS$, correct to 3 significant figures. (4 marks)



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18. In the figure, AEB is a straight line and $\triangle ABD$ is a right-angled triangle. It is given that $AC \parallel ED$ and AD is an angle bisector of $\angle BAC$.
- (a) Prove that $AE = DE$. (1 mark)
- (b) If BC is a perpendicular bisector of $\triangle BDE$ and $AD = 6$, find BD . (3 marks)
- (Leave the answer in surd form)*



19. P , Q and R are the incentre, orthocentre and circumcentre of an acute-angled triangle ABC respectively, when P is equidistant from B and C . Prove that the P , Q and R lie on the same straight line. (4 marks)